

Computer Science Graduate | Artificial Intelligence & Machine Learning Enthusiast

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GitHub: [Sabrina-Sultana13](https://github.com/Sabrina-Sultana13)
Portfolio: sabrina.vercel.app

Brac University

Bachelor's in Computer Science (B.Sc.) Dhaka, Bangladesh
Specialization: Artificial Intelligence and Machine Learning.
Relevant Coursework: Deep Learning, Data Structures & Algorithms, Software Engineering, Database Management Systems, Pattern Recognition, Computer Vision.

This section showcases hands-on application of AI/ML, Computer Vision, and Software Development processes.

Basic Emotion Detector

AI-driven real-time facial emotion recognition system using CNN and OpenCV.

- Built a real-time facial emotion recognition system using CNNs and OpenCV, achieving **92% classification accuracy** on test data.
Deployment: [Basic Emotion Detector](#)
GitHub: [Basic Emotion Detector](#)

BRACU Bhojon

Full-Stack food service/dining app

- Designed and deployed relational database architecture supporting **50+ potential users**, improving data retrieval efficiency.

Galactic Game Hub

Software Project

- Developed a centralized gaming platform integrating 7 mini-games using JavaScript, HTML, and CSS.
Deployment: [Galactic Game Hub](#)
GitHub: [Galactic Game Hub](#)

Music Cloud

Back-end relational database (PostgreSQL) implementation

- Designed robust back-end, optimized queries reducing load time by **15%**.
Deployment: [MusicCloud](#)
GitHub: [Music Cloud](#)

REI Analyst

Real Estate Investment Analysis & Financial Modeling Tool

- Built a responsive web app using JavaScript, HTML, and Tailwind CSS to automate real estate deal analysis and ROI forecasting. Deployment: [REI Analyst](#)
GitHub: [REI Analyst](#)

Real-Time Handwritten Digit Recognition

Deep Learning + Computer Vision

- Built a real-time digit recognition system using **TensorFlow** and **OpenCV** with CNN-based EMNIST inference.
GitHub: [Digit Recognition](#)

Uncertainty-Aware Federated Dual-Branch Multimodal Fusion

Privacy-Preserving Skin Lesion Diagnosis

- Designed a privacy-preserving multimodal deep learning framework for skin lesion diagnosis using federated learning. Developed a dual-branch architecture for lesion classification + segmentation and introduced DUA-D2C aggregation for heterogeneous clients. Achieved improved accuracy and robustness on HAM10000 and MILK10K datasets, ensuring safe and explainable clinical predictions. Focus: Computer Vision, Privacy, and Explainable AI (XAI) for real-world healthcare applications.

Biomedical Plant Image Segmentation using Convolutional Neural Networks

Academic Project

- **Description:** Developed a hybrid U-Net + Bi-directional Feature Pyramid Network (BiFPN) for segmenting healthy and disease-infected plant regions on PlantSeg and ISBI datasets.
Technologies: Python, TensorFlow, Keras, Matplotlib, NumPy.
Highlights: Applied data augmentation (horizontal flips, brightness adjustment), integrated multiple public datasets, conducted quantitative evaluation, ablation studies, and qualitative visualization. Ensured reproducible code for research validation.
[Project Link](#)

AI/ML & Vision: Python, TensorFlow, PyTorch, Keras, OpenCV, CNNs, Data Preprocessing, Model Training.

Programming: Java, JavaScript, TypeScript, HTML, C, SQL.

Development & Tools: Agile/Scrum, SDLC, Git/GitHub

Data & Databases: SQL (PostgreSQL/MySQL), Database Design, Power BI

Languages: Bangla (Full Professional), English (Full Professional), Hindi (Professional Working), Chinese (Limited Working), and Spanish (Limited Working).

Scout, Girls Guide, UNICEF, SNDC

Volunteer / Education Officer

- Scout (Class 3-5) – built teamwork, leadership, and community awareness.
- Girls Guide (Class 7-10) – led community service projects and social initiatives.
- SNDC (Since 2019) – Education Officer, mentoring and providing quality education to street and underprivileged students.
- Demonstrated social responsibility, leadership, and organizational skills across all roles.

- **Playing Chess:** Regularly play, sometimes participate in competitions – [Zootopiaaaa_13](#).
- Recitation :** [YouTube channel \(Sabrina\)](#).
- Other Interests:** Solving puzzles, Drawing traditional Alpona patterns, Cooking.

Referee

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